

Computer Networks Lab 05

Course: Computer Networks (CL3001) | Semester: Fall 2026

Course Coordinator: Ms. Ramsha Iqbal

Learning Objectives

- Introduction to SMTP and configuration of email services (SMTP & POP3) in Cisco Packet Tracer.
- Introduction to FTP and configuration of FTP server for file transfer between client and server.
- To observe and analyze SMTP, POP3 and FTP header format and port numbers (25, 110, 21) in simulation mode.
- Introduction to Wireshark as a packet analyzer and basic analysis of HTTP packets.

SMTP

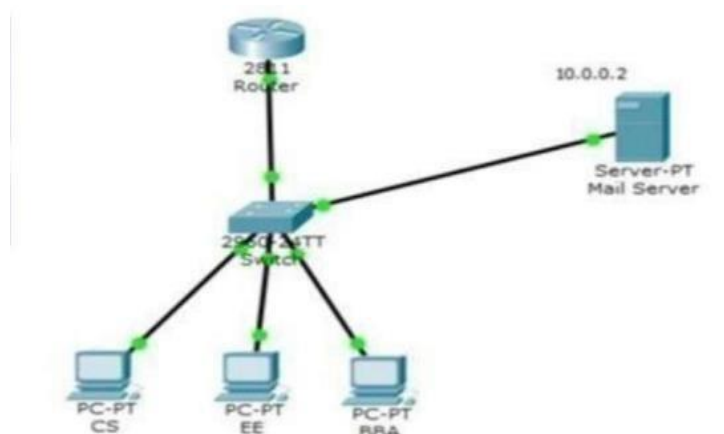
1. Introduction to SMTP

Simple Mail Transfer Protocol (SMTP) is an Internet standard for electronic mail (email) transmission. First defined by RFC 821 in 1982, it was last updated in 2008 with Extended SMTP additions by RFC 5321, which is the protocol in widespread use today. Although electronic mail servers and other mail transfer agents use SMTP to send and receive mail messages, user-level client mail applications typically use SMTP only for sending messages to a mail server for relaying. For retrieving messages, client applications usually use either IMAP or POP3.

SMTP communication between mail servers uses port 25. Mail clients on the other hand, often submit the outgoing emails to a mail server on port 587. Despite being deprecated, mail providers sometimes still permit the use of nonstandard port 465 for this purpose.

SMTP runs over TCP.

2. Implementation

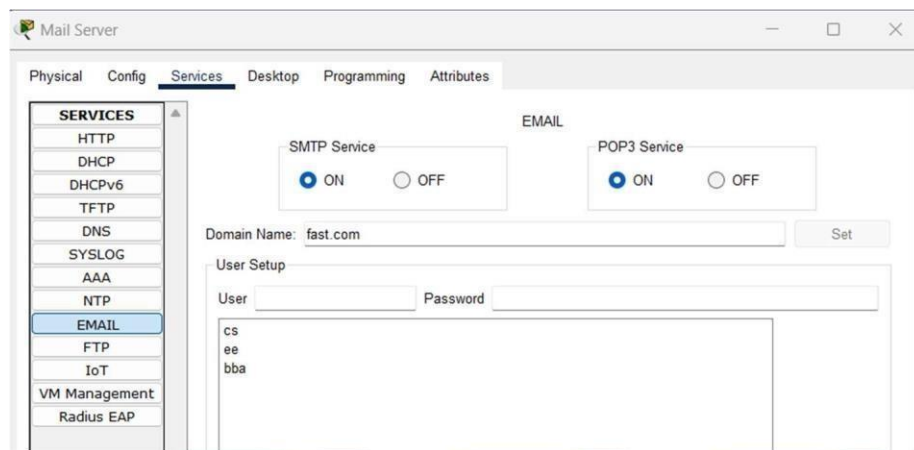


Configure and Verify Email Services:

- Click on Mail server.
- Go to services & then email services.
- Enable SMTP & POP3 Service.
- Set Domain name e.g. fast.com.

Add following users:

Username	Password
cs	123
ee	456
bba	789

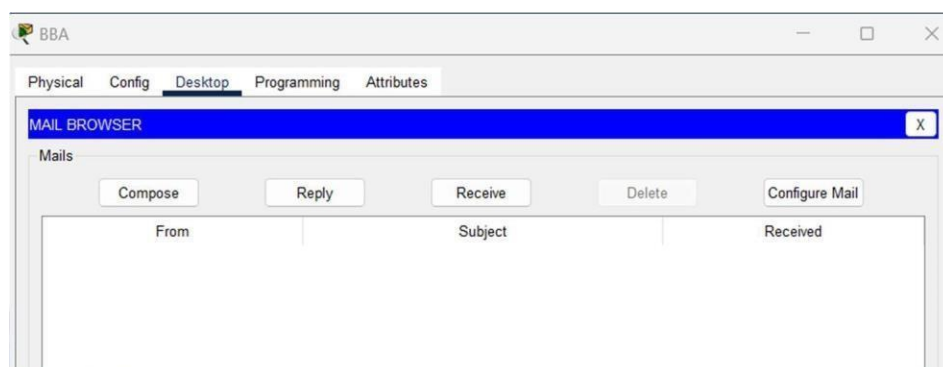
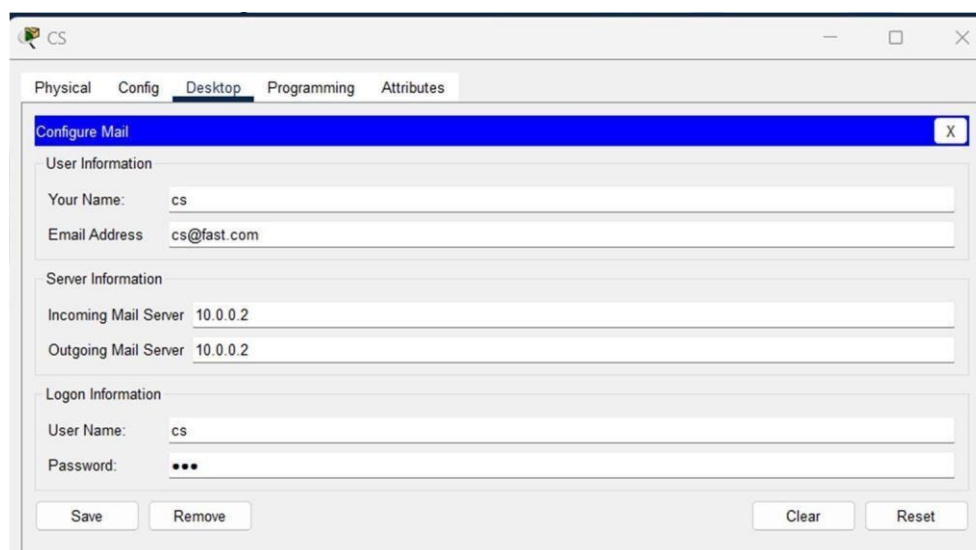


Now configure user email account.

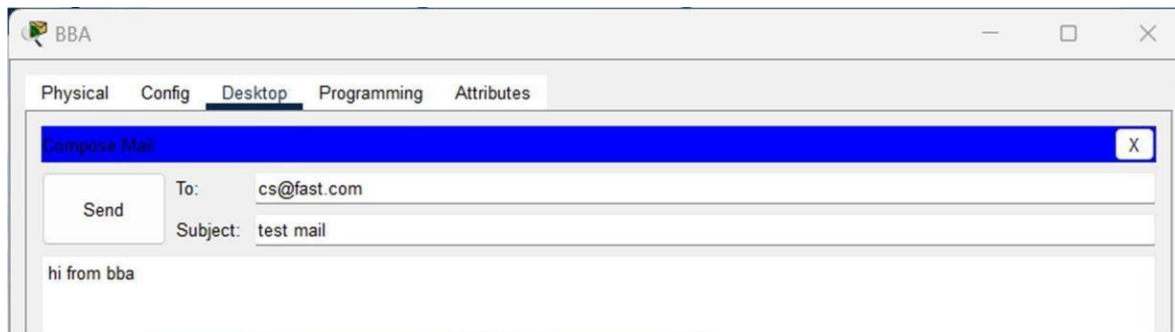
Go to PC → Desktop → Email.

Fill the following fields as shown below.

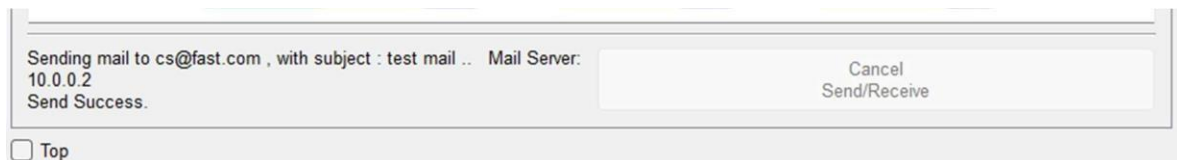
Click “Save” to save the configurations and do the same for EE and BBA.



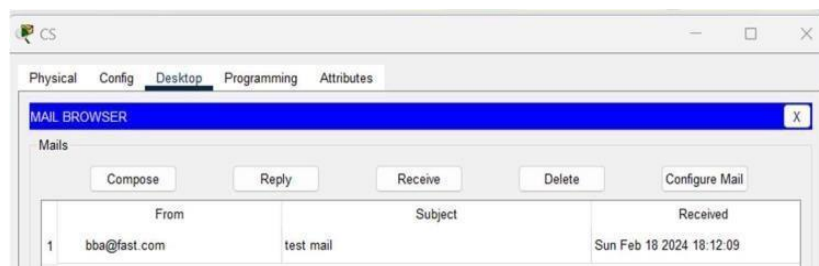
Now compose email to be sent to cs@fast.com from bba@fast.com



Click on "Send" to send Email.

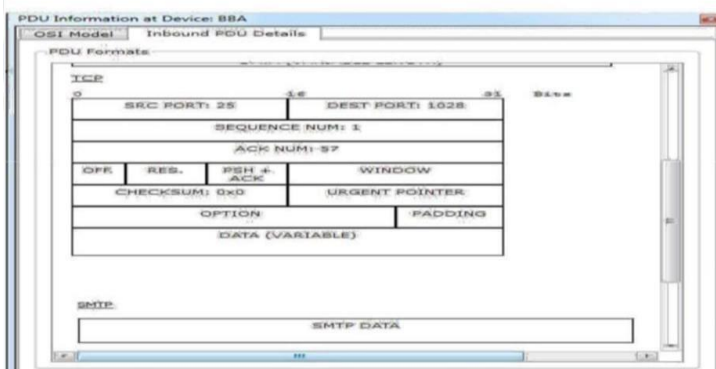
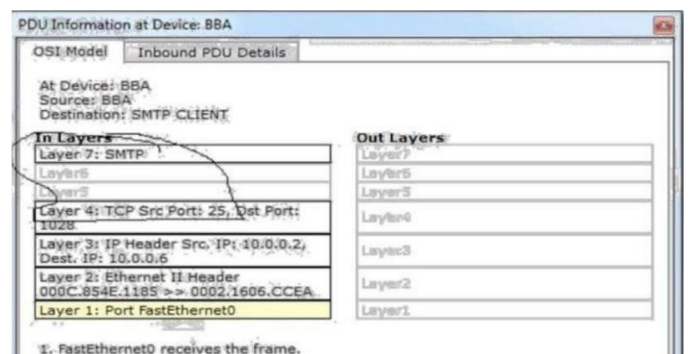
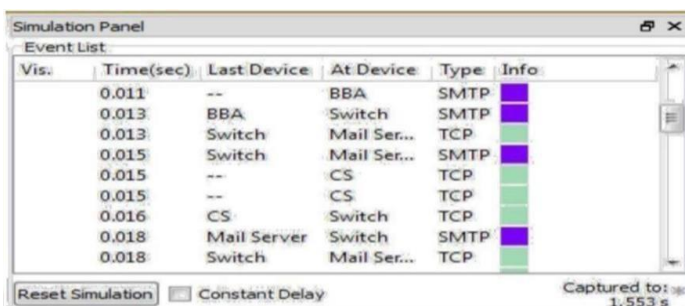


Now open email again on cs and click receive to see the email received from bba.



Simulation:

To note POP 3 header format information, go to simulation mode edit filters & check SMTP & POP 3 boxes. After that click on capture/forward button. Now see how mail server works.



FTP

1. Introduction to FTP

The File Transfer Protocol (FTP) is a standard network protocol used to transfer computer files between a client and server on a computer network. FTP is built on client-server model architecture and uses separate control and data connections between the client and the server. FTP users may authenticate themselves with a clear-text sign-in protocol, normally in the form of a username and password, but can connect anonymously if the server is configured to allow it. For secure transmission that protects the username and password, and encrypts the content, FTP is often secured with SSL/TLS (FTPS) or replaced with SSH File Transfer Protocol (SFTP). FTP uses TCP assist under layer transport protocol for data reliability transfer. It uses port 21.

FTP may run in active or passive mode, which determines how the data connection is established.

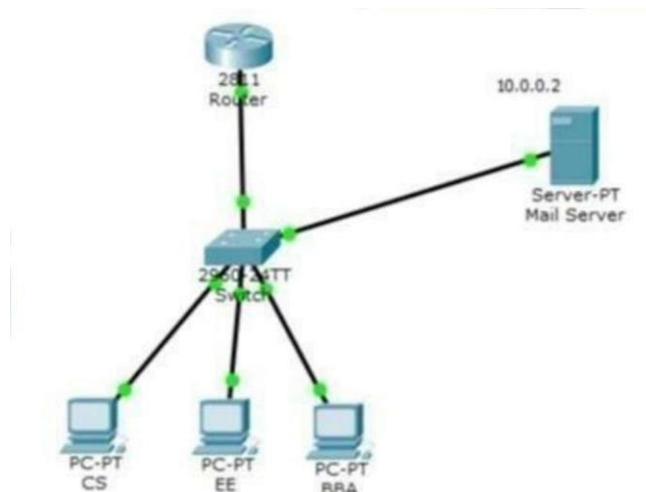
- In active mode, the client starts listening for incoming data connections from the server on port M. It sends the FTP command PORT M to inform the server on which port it is listening. The server then initiates a data channel to the client from its port 20, the FTP server data port.
- In situations where the client is behind a firewall and unable to accept incoming TCP connections, passive mode may be used. In this mode, the client uses the control connection to send a PASV command to the server and then receives a server IP address and server port number from the server, which the client then uses to open a data connection from an arbitrary client port to the server IP address and server port number received.

Both modes were updated in September 1998 to support IPv6. Further changes were introduced to the passive mode at that time, updating it to extended passive mode.

2. Implementation

In this activity, you will configure FTP server in Cisco Packet Tracer. After configuration you will transfer files between client & server. This activity is divided into 3 parts.

First Construct the topology & repeat all essential steps which we are done in pervious section.

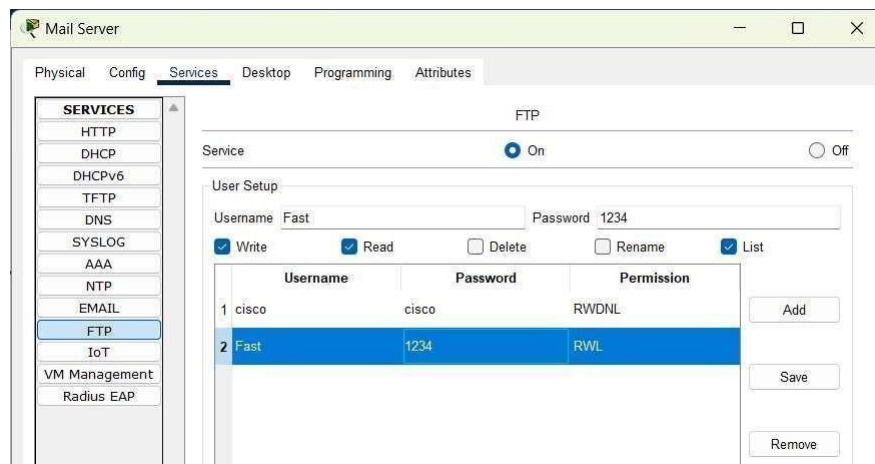


Part i: Configure FTP services on server

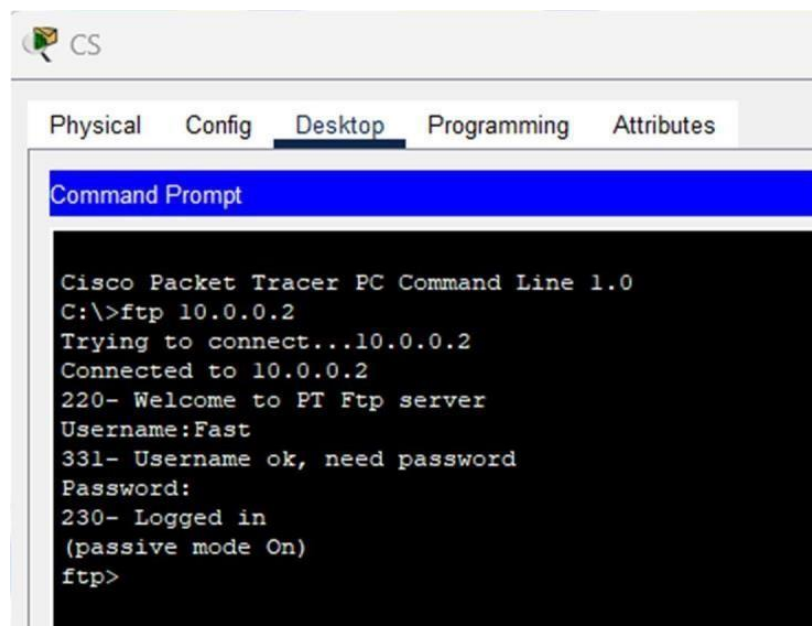
- Click server > Config tab > FTP.
- Click On to enable FTP service.

In User Setup, create the following user accounts. Click the + button to add the account:

Username	Password	Permissions
Fast	1234	Limited to Read, write & list.

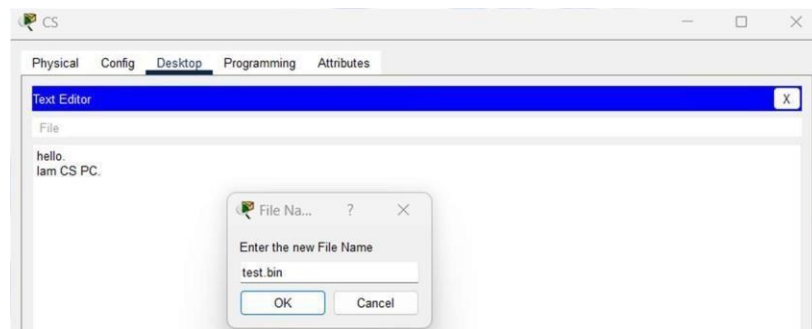


Now go to any PC Desktop command prompt. Connect with the FTP server using username & password assigned to FTP server.



Part ii: Upload file(s) to the FTP server

Go to PC Desktop text editor create file named test.bin



After creating the file go to PC Desktop command prompt and write the following command to transfer file from PC to FTP server.

put test.bin

```
ftp>put test.bin

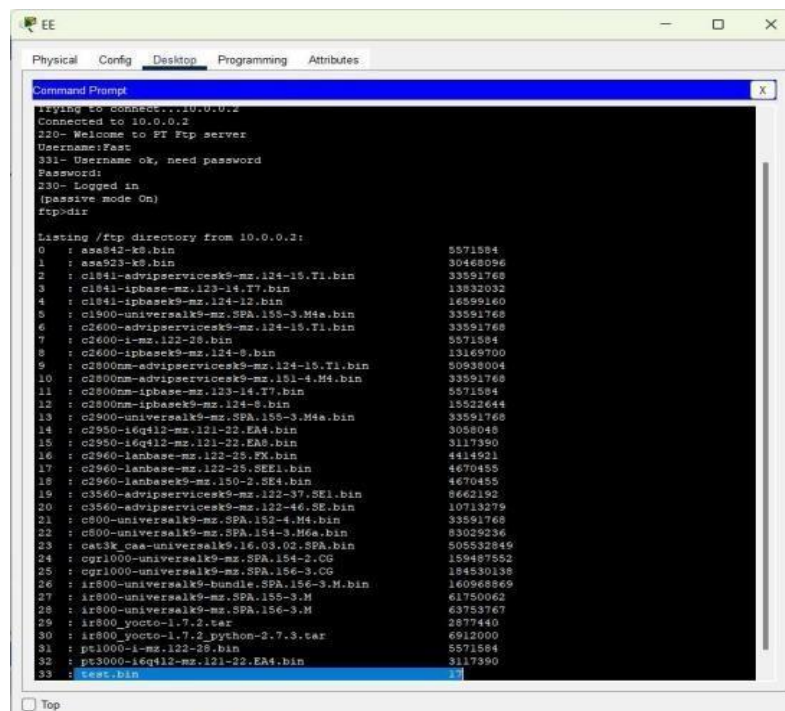
Writing file test.bin to 10.0.0.2:
File transfer in progress...

[Transfer complete - 17 bytes]

17 bytes copied in 0.037 secs (459 bytes/sec)
ftp>
```

Part iii: Upload file(s) to the FTP server

Now go to other PC desktop command prompt. Establish connection with FTP server and then write the dir command to see the files in FTP server.



Simulation:

Select the simulation mode. Go to PC desktop command prompt again make connection with FTP server using its IP address.

```

PC>ftp 10.0.0.2
Trying to connect...10.0.0.2
Connected to 10.0.0.2
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>
  
```

Now to note the FTP header format information go to simulation mode edit filters and click on FTP check box then click on capture/forward button.

How FTP server resolves the login request.

Simulation Panel

Vis.	Time(sec)	Last Device	At Device	Type	Info
	6.413	--	PC	FTP	
	6.415	PC	Switch	FTP	
	6.417	Switch	FTP Server	FTP	
	6.417	--	FTP Server	FTP	
	6.419	FTP Server	Switch	FTP	
	6.421	Switch	PC	FTP	
	6.441	--	PC	TCP	
	6.442	PC	Switch	TCP	
	6.444	Switch	FTP Server	TCP	

FTP

220

Welcome to PT Ftp server

FTP

USER

cisco

FTP

331

Username ok, need password

FTP

PASS

cisco

FTP

230

Logged in

PDU Information at Device: PC

OSI Model Inbound PDU Details

At Device: PC
Source: FTP Server
Destination: 10.0.0.2

In Layers

Layer 7: FTP

Layer6

Layer5

Layer 4: TCP Src Port: 21, Dst Port: 2029

Layer 3: IP Header Src. IP: 10.0.0.2, Dst. IP: 10.0.0.3

Layer 2: Ethernet II Header 000C.854E.1185 >> 0001.96A6.CD3B

Layer 1: Port FastEthernet0

Out Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer2

Layer1

1. FastEthernet0 receives the frame.

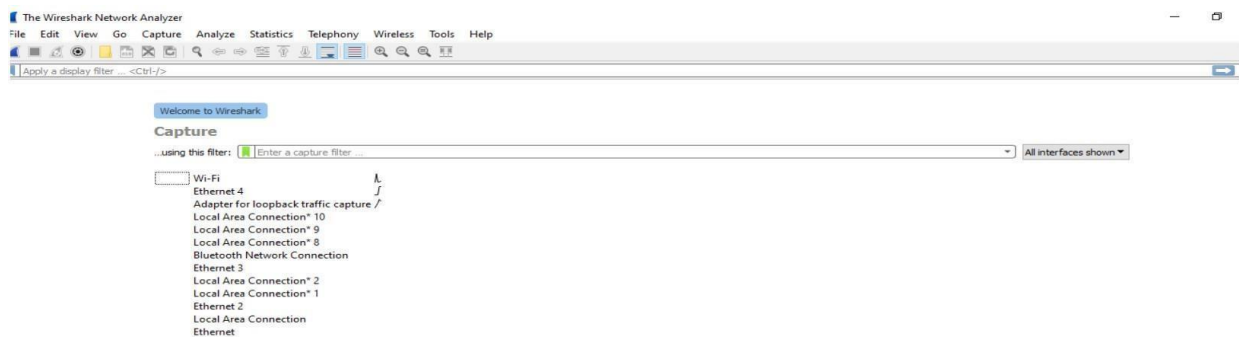
Wireshark

Wireshark is a free and open-source packet analyzer. It is used for network troubleshooting, analysis, software and communications protocol development and education.

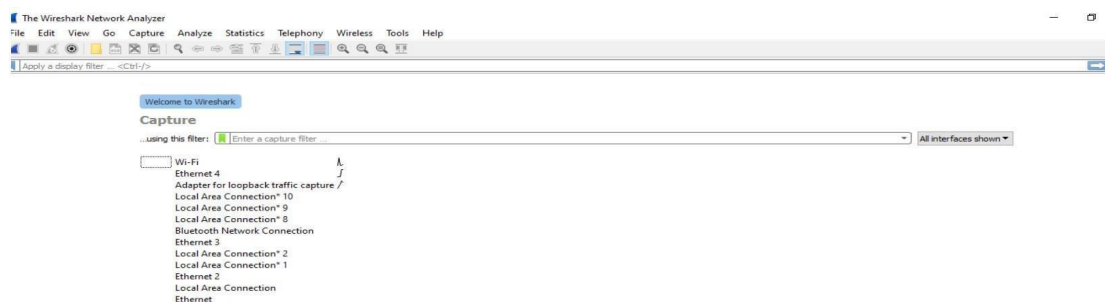


Why we use Wireshark?

Wireshark has many uses, including troubleshooting networks that have performance issues. Cybersecurity professionals often use Wireshark to trace connections, view the contents of suspect network transactions and identify bursts of network traffic.

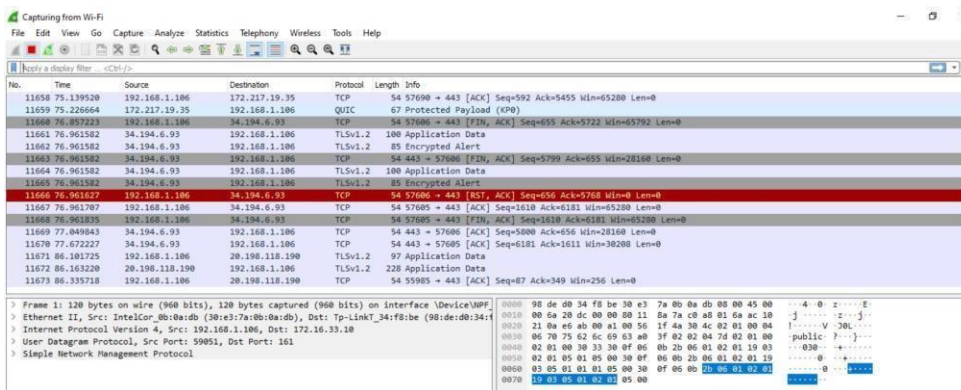
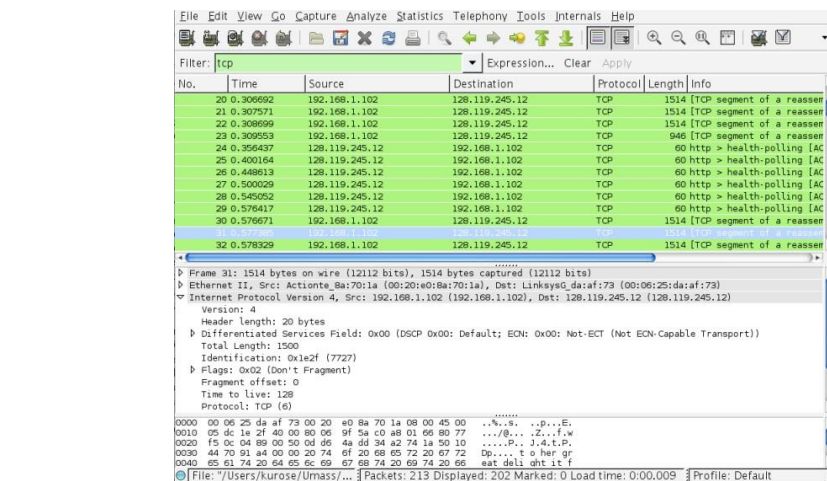


Open Wireshark



Select the technology you used for packet analysis

First, filter the packets displayed in the Wireshark window by entering “tcp” (lowercase, no quotes, and don’t forget to press return after entering!) into the display filter specification window towards the top of the Wireshark window.



HTTP Packet Analysis

Goto website below:

<http://testphp.vulnweb.com/login.php>

Username: test

Password: test

Let’s take a look here what we can understand from here:

